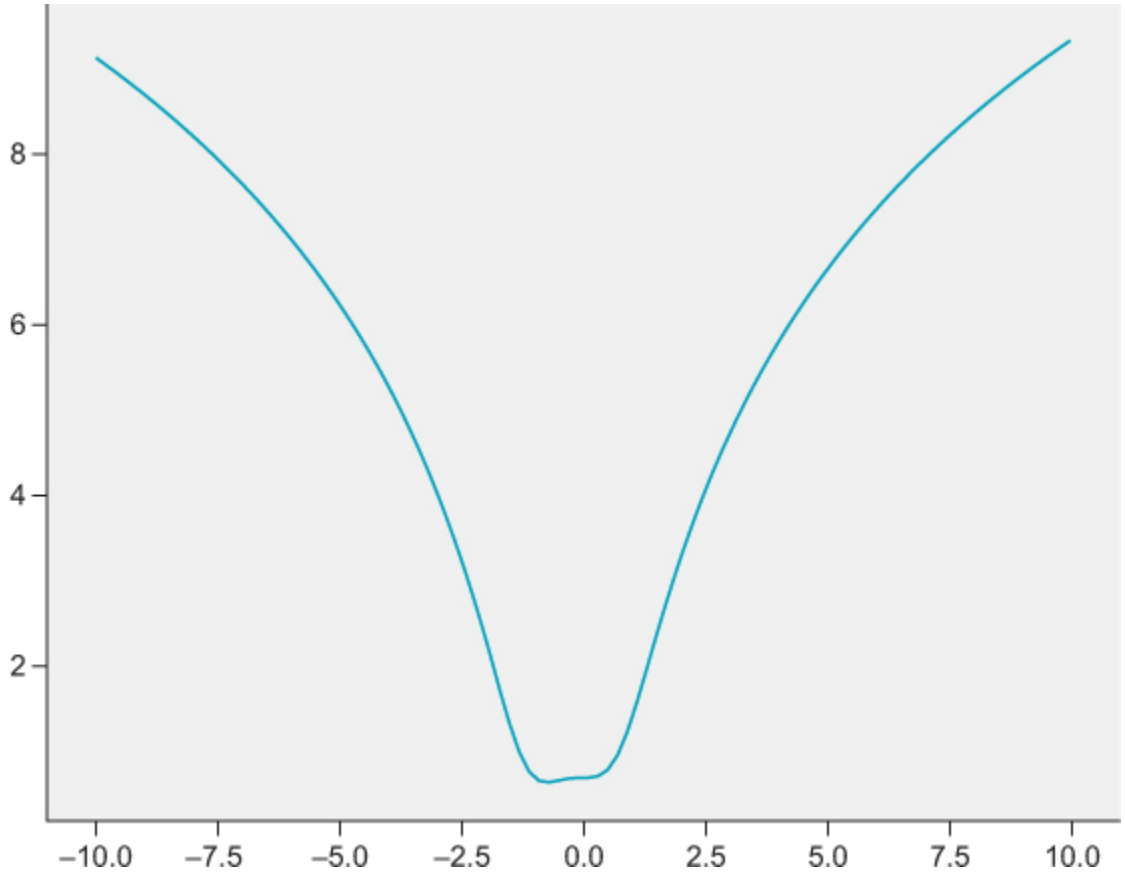
Figure A.1. The output of a simple compositional function, $f(x) = \log(x^4 + x^3 + 2)$



The chain rule in calculus gives us a way of computing derivatives of compositional functions by decomposing them into pieces. If you've heard of *backpropagation*, it's basically just the chain rule applied to neural networks with some tricks to make it more efficient. For our example case, let's rewrite the previous function as two functions:

$$h(x) = x^4 + x^3 + 2$$
$$f(x) = \log(h(x))$$

We first compute the derivative of the "outer" function, which is $f(x) = \log(h(x))$, but this just gives us df/dh and what we really want is df/dx. You may have learned that the derivative of natural-log is

$$\frac{d}{dh}\log h(x) = \frac{1}{h(x)}$$